

ALY 6060 Decision Support and Business Intelligence

Module 1 – Analytics and Influencing

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PART 1

- a. **Question 1** - Do you agree or disagree with the idea of analytics being a “people business” at its core? Why or why not?

The notion that analytics is fundamentally a "people business" is compelling and has merit. I agree with this perspective, as analytics, regardless of the complexity of its tools or techniques, is ultimately intended to serve human decision-making. This report delves into why analytics can be considered a people-centered business and examines whether its core purpose is indeed to influence decisions or if there may be other ultimate objectives.

At its core, analytics is a field driven by the need to understand and interpret data for organizational benefit. Business Intelligence (BI), a critical component of analytics, aims to guide organizations toward informed decision-making and strategic planning. Business Intelligence helps “employees or individuals... set the company’s direction and strategy” and spot “opportunities... to improve [the organization’s] competitive advantage in the industry” (Importance of Business Intelligence – video NEU). The concept of analytics as a "people business" implies that the end goal of data insights is not merely to identify trends but to make those trends actionable for the benefit of stakeholders, including executives, managers, and employees.

Analytics professionals work in close collaboration with organizational decision-makers, helping them understand the numbers in a way that aligns with business goals. In line with the idea that BI provides insights into “present realities of the market or the environment” (Importance of Business Intelligence – video NEU), data analysis supports human insights and foresight, enabling leaders to act upon real-time data or forecasted trends. This “people business” focus underscores the fact that analytics is not an isolated activity; it is embedded within the organization’s broader ecosystem where humans interpret and act on the data.

- b. **Question 2** - Is the ultimate goal of analytics to influence decision-making? If so, how does that work? If not, what do you think the ultimate goal of analytics is?

The importance of analytics in influencing decisions cannot be overstated. Analytics provides a way to sift through vast amounts of disorganized data, creating “a clean, focused data set that is easy for everyone else to follow” (Importance of Business Intelligence – video NEU). By translating complex datasets into actionable insights, analytics professionals ensure that data-driven recommendations align with organizational objectives. Business Intelligence helps in

creating “actionable options” that allow stakeholders to explore “where we can improve” by comparing internal performance against industry standards.

Moreover, analytics supports the cyclical process of questioning, analyzing, and refining, known as the cycle of analytics, which reflects the ongoing need to answer emerging questions and iterate on previous analyses. This process serves decision-making at every level, as BI tools allow organizations to react to changing questions and expectations through repeated data access, exploration, and sharing. In this way, analytics not only facilitates one-time decisions but fosters continuous improvement by enabling iterative, adaptive decision-making.

Some may argue that analytics serves a broader purpose beyond immediate decision-making, focusing instead on driving efficiency, identifying trends, and creating a competitive advantage. “Each Business Intelligence tool plays a supporting role in one or more phases of the decision-making cycle. “ (Passion Group - List of BI tools 2024- Decision making process).

In this view, the ultimate goal of analytics is to provide a framework for understanding organizational performance and the market environment. For example, analytics enables organizations to spot hiring needs, assess market positioning, and even detect patterns that would otherwise remain hidden.

Analytics can lead to strategic shifts such as cost-cutting measures, competitive analysis, investment strategy, and merging risks. These objectives may not result in immediate decisions but instead build a foundation for sustained, data-driven strategy over time. Data science techniques, such as predictive modeling and machine learning, allow organizations to forecast future patterns, which can create a proactive rather than a reactive stance in a business’s strategic planning. According to Gartner, business analytics involves “data mining, predictive analytics, applied analytics, and statistics,” all of which serve as tools to build a more comprehensive understanding of an organization’s operations (Tableau - How BI, data analytics, and business analytics work together).

In conclusion, analytics can be seen as both a “people business” and a decision-support tool. The ultimate goal of analytics may indeed be to influence decision-making, but this process relies on an analyst’s ability to present data in a manner that is both accessible and actionable for stakeholders. The influence of analytics on decision-making is most effective when analysts work in close collaboration with business leaders, ensuring that data insights align with organizational

goals and strategies. However, analytics also serves broader objectives by enhancing efficiency, supporting strategic planning, and building a data-informed culture across the organization.

Analytics goes beyond influencing individual decisions—it creates a continuous improvement cycle, where data insights lead to questions, exploration, and deeper organizational learning. This dual function of analytics underscores its value as both a decision-making aid and a catalyst for long-term organizational insight. In this sense, analytics is not merely a tool; it is an integral part of an organization's efforts to remain competitive and agile in an increasingly data-driven world.

PART 2

The dashboard created presents a detailed representation with multiple visualizations of stock performance data for five major companies, focusing on their average prices, adjusted closing prices, and trading volume across 2022. Each graph depicts an analysis of whether stock splits may be appropriate for each company. (appendix) (dataset chosen from Yahoo Finance)

a. Dashboard Explanation:

1. Average Highest and Lowest Closing Prices in 2022 (Left Bar Chart):

High Prices: The bar chart at the top shows the average highest closing prices for each company in 2022. Tesla and Microsoft exhibit the highest values, while Amazon (AMZN) and Google are in the middle, and Apple has the lowest average high.

Low Prices: The lower bar chart shows the average lowest closing prices for each company. Similar to the high prices, Tesla and Microsoft maintain higher average lows, while Apple, Amazon, and Google are lower.

2. Adjusted Closing Price of Stocks on January 3, 2022 (Line Chart):

This line chart tracks the adjusted closing prices for each company over the quarters (Q1 to Q4) in 2022. There's a general decline in adjusted closing prices for most companies, with Tesla and Microsoft showing more significant drops.

3. Volume of Shares Traded in 2022 (Pie Chart):

This pie chart illustrates the average volume of shares traded for each company in 2022. Apple and Tesla account for the largest portion of trading volume, while Amazon, Google, and Microsoft have smaller proportions.

4. Trading Volume Comparison (Bar Chart):

This bar chart compares the monthly trading volume across all five companies throughout the year. Trading volume fluctuates, with some spikes, especially for Apple and Tesla. The bar colors correspond to each company's ticker.

5. Trend Analysis of Shares (Small Line Charts):

These mini line charts track the trend over time for average volume, high, low, and close prices. The red lines indicate overall trends, and for most companies, there's a declining trend in the average high, low, and close prices by the end of 2022.

b. Stock Split Analysis

To decide whether the president of each company should consider a stock split, we'll consider two primary factors from the data, High Share Prices where companies with higher share prices may choose to split to make shares more affordable to retail investors and high trading volume often suggests strong investor interest, which might support a stock split to increase accessibility.

c. Recommendations for Each Company

1. Apple:

Apple's stock has the lowest average high price among the five companies, however, Apple has one of the highest trading volumes, showing strong demand.

A split may not be necessary, as Apple's share price is already relatively accessible. However, high trading volume could justify a split if the company wants to attract even more retail investors.

2. Amazon:

Amazon's average price is in the middle range and has a moderate trading volume compared to Apple and Tesla.

Amazon could consider a stock split if they wish to increase liquidity and make shares more attractive to retail investors, but it may not be a high priority given the moderate volume.

3. Google:

Google's average prices are in the lower-middle range, but, it has lower trading volume compared to Apple and Tesla.

A split could make Google's stock more attractive to smaller investors, but given the lower trading volume, the impact might be limited.

4. Microsoft:

Microsoft has high average prices, second only to Tesla and has lower trading volume than Apple and Tesla.

A stock split could make Microsoft's shares more accessible, especially given the higher price level. This might increase investor interest and trading volume.

5. Tesla:

Tesla has the highest average prices among the five and also has a very high trading volume.

Tesla should consider a stock split. The high price and trading volume indicate strong investor demand, and a split would make the shares more affordable, likely sustaining or even increasing investor interest.

d. Conclusion:

Tesla and **Microsoft** are strong candidates for stock splits based on their high share prices. Their presidents should consider a split to enhance share accessibility and maintain investor interest.

Amazon and **Google** may consider a split but are not as high-priority candidates due to moderate trading volumes and middle-range prices.

Apple may not need a split given its already accessible share price, though high trading volume suggests there could still be benefits in doing so.

References

Part 1:

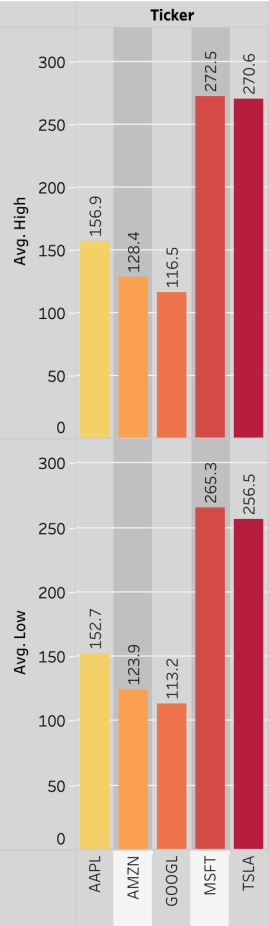
- a. NEU Module 1 – Lesson 1-1 – Business Intelligence – Video: Importance Of Business Intelligence - https://northeastern.instructure.com/courses/196473/pages/lesson-1-1-business-intelligence?module_item_id=11159992
- b. Tableau article - <https://www.tableau.com/business-intelligence/what-is-business-intelligence>
- c. Passioned Group - <https://www.passionned.com/business-intelligence/tools/>

Part 2:

- a. Tableau for Dashboard creation: <https://www.tableau.com>
- b. Yahoo Finance: <https://finance.yahoo.com>

APPENDIX

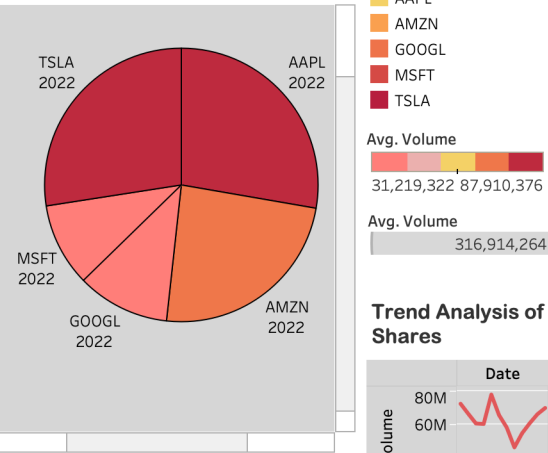
Avg Highest and Lowest closing price in 2022



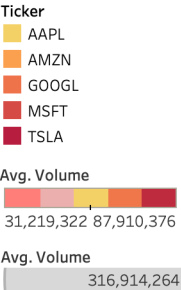
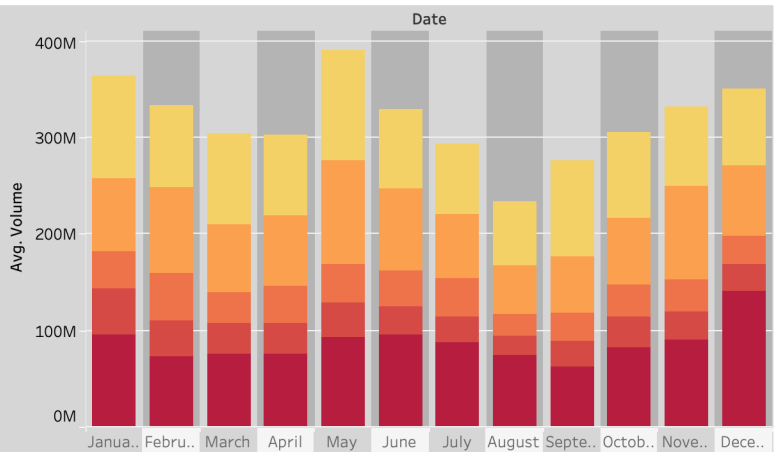
Adjusted closing price of stocks on January 3, 2022.



Volume of Shares traded in 2022



Trading Volume Comparison



Trend Analysis of Shares

